Decision-Chain Alignment: Method Comparison

Steering a frozen 12L-8H-512D GPT-NeoX toward decision function f

Setup

A pretrained model picks decision function f vs. g at each chain step $\approx 50/50$. Each method tries to push the model toward always picking f. Two evaluation slices:

- **val_full** — mixed-pattern val (the honest test for off-distribution alignment).
- **val_ffff** — all-f val (in-distribution sanity check).

Metrics: **f-sel** = per-step fraction picking f's letter %; **Full align** = strict per-example all-f %; **Op-acc** = arithmetic correctness of generated blocks %.

Table 1: val_full (mixed-pattern). Δ = change vs. baseline in percentage points. *Memory alignment matches or beats every finetuning baseline on alignment while preserving arithmetic best.*

Method	**f-sel** (%)	**Full align** (%)	**Op-acc** (%)
Baseline (no training)	52.5	8.8	94.8
GRPO (chain reward)†	60.2 +7.7	11.4 +2.6	93.9 −0.9
LoRA, mode A	81.6 +29.1	37.2 +28.4	94.3 −0.5
LoRA, mode B	96.6 +44.1	83.5 +74.6	89.3 −5.5
Naive FT (full), mode A	96.0 +43.5	84.2 +75.3	92.6 −2.1
Naive FT (full), mode B	92.9 +40.5	69.8 +60.9	84.7 −10.0
Memory align (L=2, N=64)	**97.6** +45.1	**88.0** +79.2	**94.1** −0.7

Table 2: val_ffff (all-f, in-distribution sanity check). Every aligning method saturates near the ceiling.

Method	**f-sel** (%)	**Full align** (%)	**Op-acc** (%)
Baseline (no training)	76.8	64.8	95.1
GRPO (chain reward)†	86.8 +10.0	76.8 +12.0	97.2 +2.1
LoRA, mode A	99.3 +22.5	98.2 +33.4	99.8 +4.7
LoRA, mode B	100.0 +23.2	100.0 +35.2	99.9 +4.8
Naive FT (full), mode A	99.9 +23.1	99.8 +35.0	99.7 +4.6
Naive FT (full), mode B	99.7 +22.9	99.4 +34.6	99.4 +4.3
Memory align (L=2, N=64)	99.8 +23.0	99.3 +34.5	99.7 +4.6

† GRPO evaluated at n_{eval}=500; all other rows at n_{eval}=2000.

Mode A vs. B

- **Mode A** — ffff-only data, GT labels (standard LM loss).
- **Mode B** — mixed data with f-override at decision points (the same alignment signal Memory uses).

Reading the results

1. **GRPO is the wrong tool for alignment.** GRPO optimizes for chain correctness, not for which decision function gets picked. Mixed-val f-selection moves only +7.7 pp; full-alignment +2.6 pp. Every other aligning method exceeds +28 pp on f-selection.
2. **Memory alignment dominates the alignment $\times$ arithmetic trade-off.** Best f-selection (97.6 %) and best full-alignment (88.0 %) on mixed val, with the smallest op-accuracy regression (-0.7 pp). And it does this with $\sim$328K trainable params — about 1% of the base model.
3. **Naive FT mode B exhibits catastrophic forgetting** (-10 pp op-accuracy). With 37.9M free parameters, the override-on-mixed-data signal warps arithmetic. LoRA at the same supervision stays stable because the adapter only moves the model in a low-rank subspace.
4. **LoRA mode A generalizes poorly off-distribution.** It saturates on ffff val (98.2 % full align) but only reaches 37 % on mixed val — a 61 pp gap that exposes the limit of training on ffff-only data.